YUVRAJ RUPAREL

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EDUCATION

Duke University **08/2025 – 05/2029**
Bachelor of Science in Mechanical Engineering *Durham, North Carolina*
- GPA: 4.0, Dean's List, Activities and Societies: Duke Robotics Club

Aditya Birla World Academy **08/2013 – 05/2025**
- IBDP: 40/45, 4 HL 2SL, Activities: Math Club President, Chanakya (Yellow) House Captain *Mumbai, India*

Johns Hopkins CTY, Summer School: Year 1: Arduino Prototyping, Year 2: Investigations in Engineering **05/2021 & 06/2022**

PROFESSIONAL EXPERIENCE

Duke Robotics Club **01/2026 – Present**
Electricals: PCB Designer
- Designed a Printed Circuit Board (PCB) in KiCad that integrates a custom built in voltage sensor, temperature – humidity sensor, I2C pressure sensor with bidirectional level shifter for 5.5V→3.3V logic translation, and Arduino Nano for real time data logging.
- Integrated PCB into two current underwater autonomous robots, Oogway and Crush, for RoboSub.

Mumbai University **07/2022 – 09/2024**
Research Intern: Worked on 3 Published Projects *Mumbai, India*
- Synthesis of Ni-BCA/CF Hybrid Composite for high performance Li-ion Battery Anode: Engineered a nickel based composite achieving capacity with exceptional rate of performance retention (54%) at high charging speeds.
- Synthesis of Ag-Bimetallic Antimicrobial Nanoparticle Hydrogel for Burn Relief: Green-synthesized silver nanoparticles with antimicrobial properties; hydrogel demonstrates 110% water absorption for advanced wound healing applications.
- Biodegradable Aqua balls for Plant Hydration & Nutrition: Developed water absorbing polymer spheres that absorb 248 mL of water per 2 grams and completely biodegrade within seven weeks which are used for smart plant irrigation.

ThyssenKrupp **12/2023**
Engineering Intern *Mumbai, India*
- Assisted with CAD Modeling to design elevators. Conducted physical simulations on parts to test quality of components.
- Documented the process, including writing reports on the simulations. Cleaned and analyzed data on profit and revenue.

TECHNICAL PROJECTS

Spiderbot Mark 1 - Simulated Quadruped Robot **08/2026 – Present**
- Designed a 1.1kg, 2 DOF per leg quadruped robot in Fusion with 8 ST3215-HS servos and Raspberry Pi 5. click here
- Built MuJoCo simulation showing crawl gait for forward/backward translation & clockwise/anticlockwise rotation.
- Mk 2: RL trained gait, Mk 3: Sensors & Electronics (PCB), Mk 4: WiFi & Bluetooth, Mk 5: Camera (CV), Mic, & Voice

Humidifier and Dehumidifier for Duke Phytotron Building **08/2025 – 12/2025**
- Engineered a Humidifier and Dehumidifier for R2 Chambers. Interfaced system with TC2 microcontroller using ESP32.
- Provided CAD files, engineering drawings, circuit schematics, and 1 prototype for staff to replicate for 15 more chambers.

Automated Waste Segregating Dustbin **04/2024 – 11/24**
- Developed and Co-authored a research paper in SciEP: click here
- Prototyped and built an AI powered Smart Dustbin for efficient and automated waste segregation, capable of identifying 3 different types of waste and segregating them using a conveyor belt and a sorting system.
- Installed in 3 residential societies in Mumbai, sorting over 30kg of waste in each complex.

Aramco F1 in Schools Competition **12/2021 – 09/2023**
Team Leader and Lead Design Engineer - 1st Rookie team to represent Team India in World Finals *Singapore*
- 3D modelled and manufactured miniature F1 Car to compete. Wrote an Engineering Portfolio with car details and drawings.
- Regional Round: Won Best Enterprise Portfolio, Best Rookie Team, and Regional Champions.
- National Round: Won Best Scrutineering Award, National Collaboration Award, People's Choice Award, Second Runners Up
- World Finals: Won Chair of Judges Recognition of Achievement Award, sponsored by Redbull Racing.

ACADEMIC AWARDS AND HONORS

- Johns Hopkins CTY Honor: Recognized as "one of the brightest students in the world", scoring top 20% from 19,000 students across 84 countries for CTY 2020 – 2021 Talent Search **08/2021**
- IRIS National Fair: Selected for National Round as the only Chemistry Research Project across India **10/2023 – 02/2024**

Crest Gold Awards
- 4 awards: Ag-Ni Hydrogel (05/2024), Ni-BCA/CF Hybrid Composite (08/2024), AI Powered Smart Dustbin (10/2024), Biodegradable Aqua Balls (12/2024)

TECHNICAL SKILLS

- **Programming Languages:** Python, C++, HTML, CSS, SQL
- **Technologies:** Git/Github, VS Code, Solidworks, KiCad, Altium, Autodesk Fusion, AutoCAD Onshape, Keyshot
- **Languages Spoken:** English, Spanish, Hindi, Gujarati, Marathi, Kutchi